



GEMINI ENTERPRISE **Global Hackathon**

The Team:

PRAHARI

presents

SecOps Nexus

Secure AI access to enterprise security data via MCP + RBAC

Team members:

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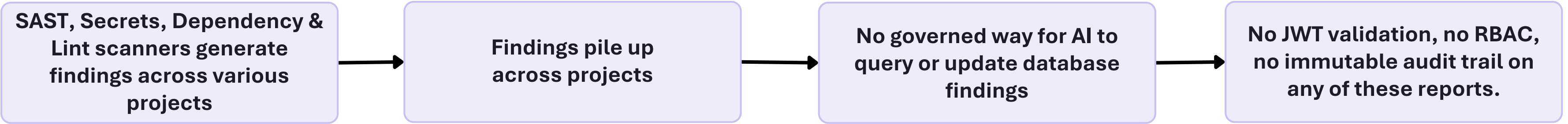
Sayonika Pal

Shaily Priya

Subhashree Prusty

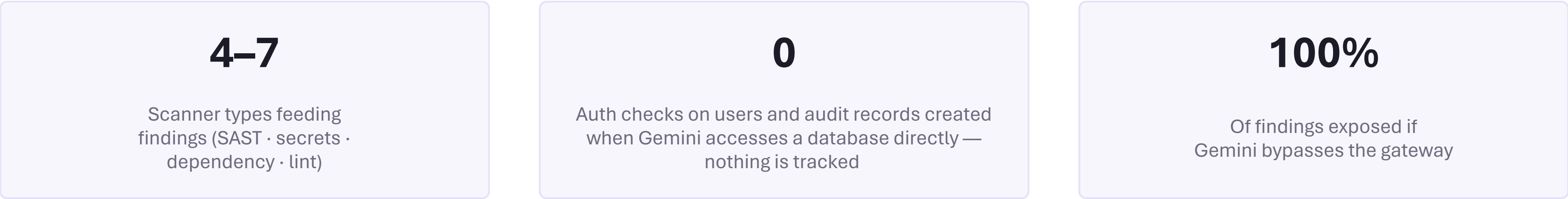
The Problem:

Security AI Without Guardrails Is a Liability



DB ACCESS WITHOUT GOVERNANCE = SECURITY LIABILITY

Security findings from various security scans pile up across projects with no safe way for an AI assistant to query, reason over, or update them. Giving Gemini direct database access means an AI that can read everything can also leak credentials, bypass role boundaries, and leave no trace of what changed. Hence, enterprises need a governed gateway, one that validates every identity, enforces role-based access before any query reaches the database, and writes an immutable record on every status change.



The Solution

SecOps Nexus is a connector that closes the governance gap between Gemini and enterprise data.

It is a production-ready MCP connector that sits between Gemini Enterprise and your security findings database, enforcing JWT authentication, server-side RBAC, and immutable audit logging on every single request, before any data ever reaches the AI.

01**REQUEST**

user gives a prompt in Gemini Enterprise chat, selects the right MCP tool and sends a Streamable HTTP call to /mcp on Cloud Run

02**AUTHENTICATE**

JWT validated on every request and the user

03**RESOLVE**

resolves identity via DB lookup for seeded users or email mapped to role via USER_ROLE_MAP in Secret Manager

04**AUTHORIZE**

checks the RBAC matrix with VIEWER read-only, SECURITY_ENGINEER read + update, SECURITY_ADMIN full access

05**EXECUTE**

runs the operation for writes, an atomic transaction updates findings and inserts into audit_logs

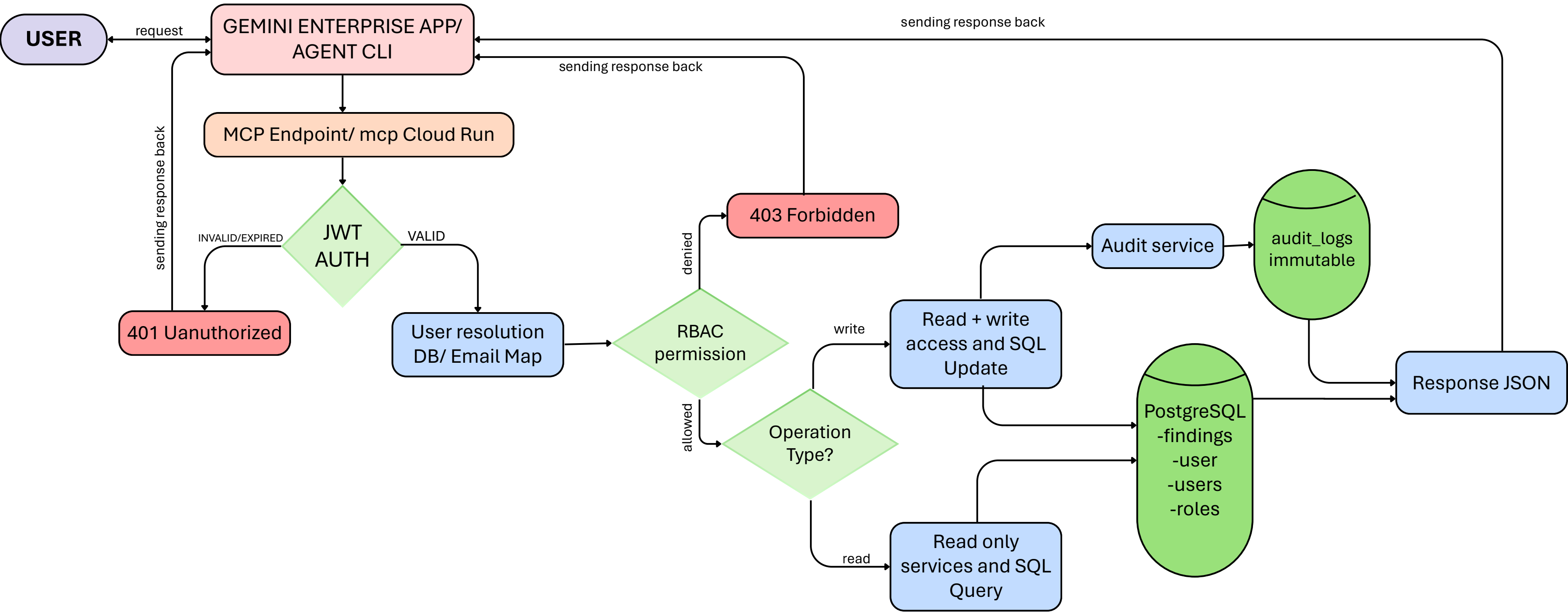
06**RESPOND**

Structured JSON returned to Agent and then structured result returned to user, audit record committed

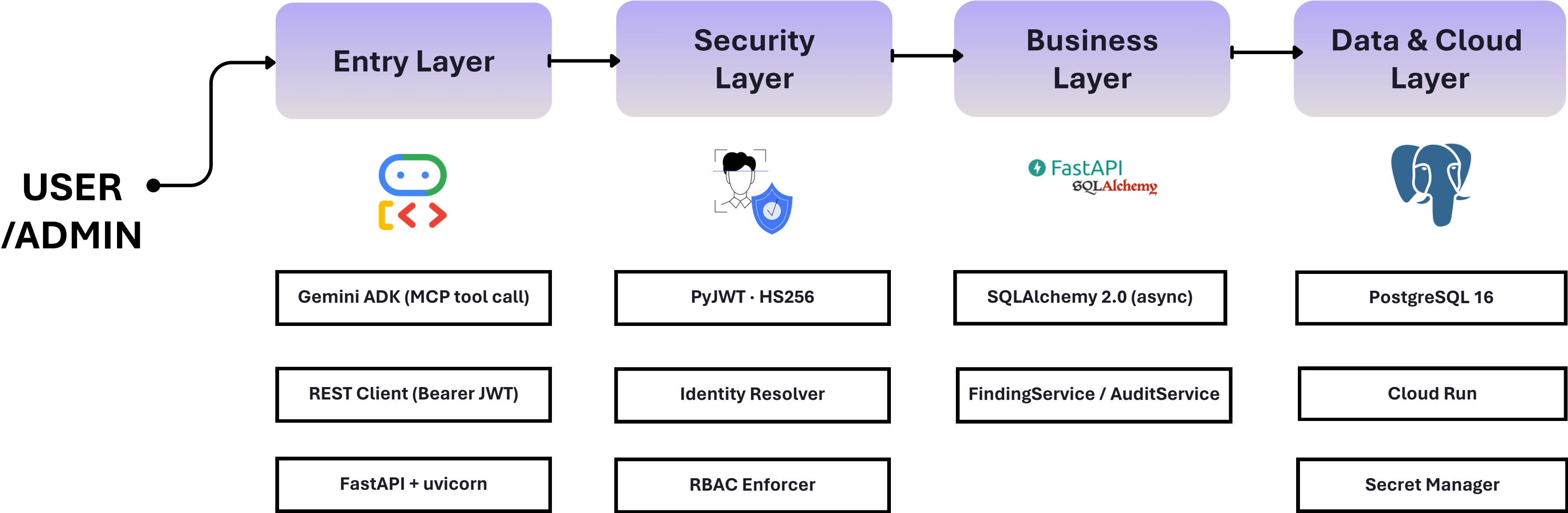
Key Differentiators:

- Governed, not just connected — auth + RBAC + immutable audit on every call
- Two entry points, one security layer (MCP tools and REST API share it)
- Two identity paths — seeded demo users and an enterprise email-role map
- Standalone and reusable by any Gemini ADK app, MCP client, or REST consumer

The Project Architecture

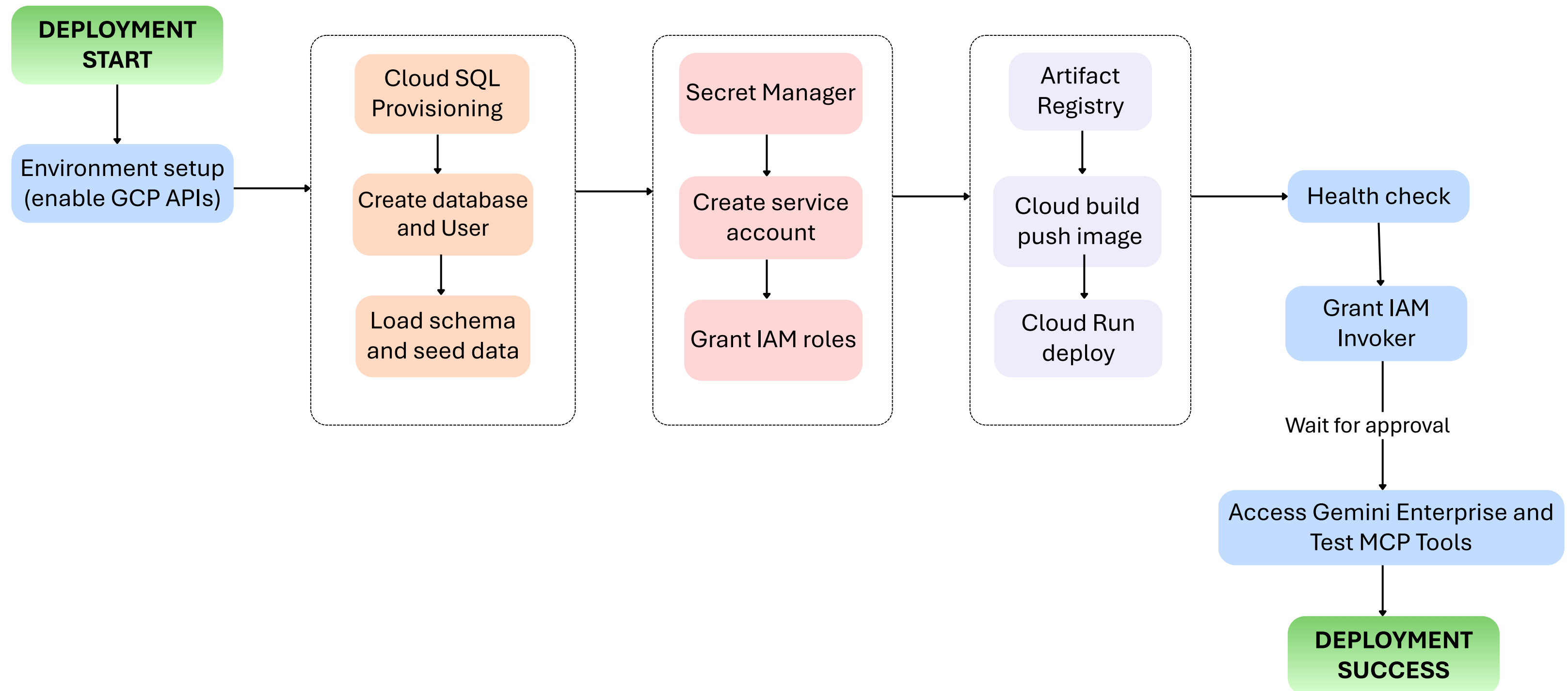


Tech Stack Workflow



Every request — MCP tool call or REST call — passes through the Security Layer before it ever reaches business logic or data.

The Deployment Architecture



Next Steps

From Prototype → Enterprise Platform

NOW – Core loop

- 5 MCP tools live on Cloud Run
- 6 REST API endpoints on Cloud Run
- JWT auth Security enforced on every request
- 3 RBAC roles (Viewer/Engineer/Admin)
- Immutable audit trail
- Vertex AI Agent Engine

NEXT – Enterprise integration

- Real scanner ingestion and live findings directly from security scanners via webhook or CI/CD pipeline trigger instead of seeded SQL
- Multi-tenant projects by extending the data model
- OAuth / SSO identity
- GitLab / GitHub webhooks integrations.

SCALE – Enterprise security

- Multi-tenant RBAC
- SRE real-time notifications
- SLA tracking
- OpenID Connect / SSO

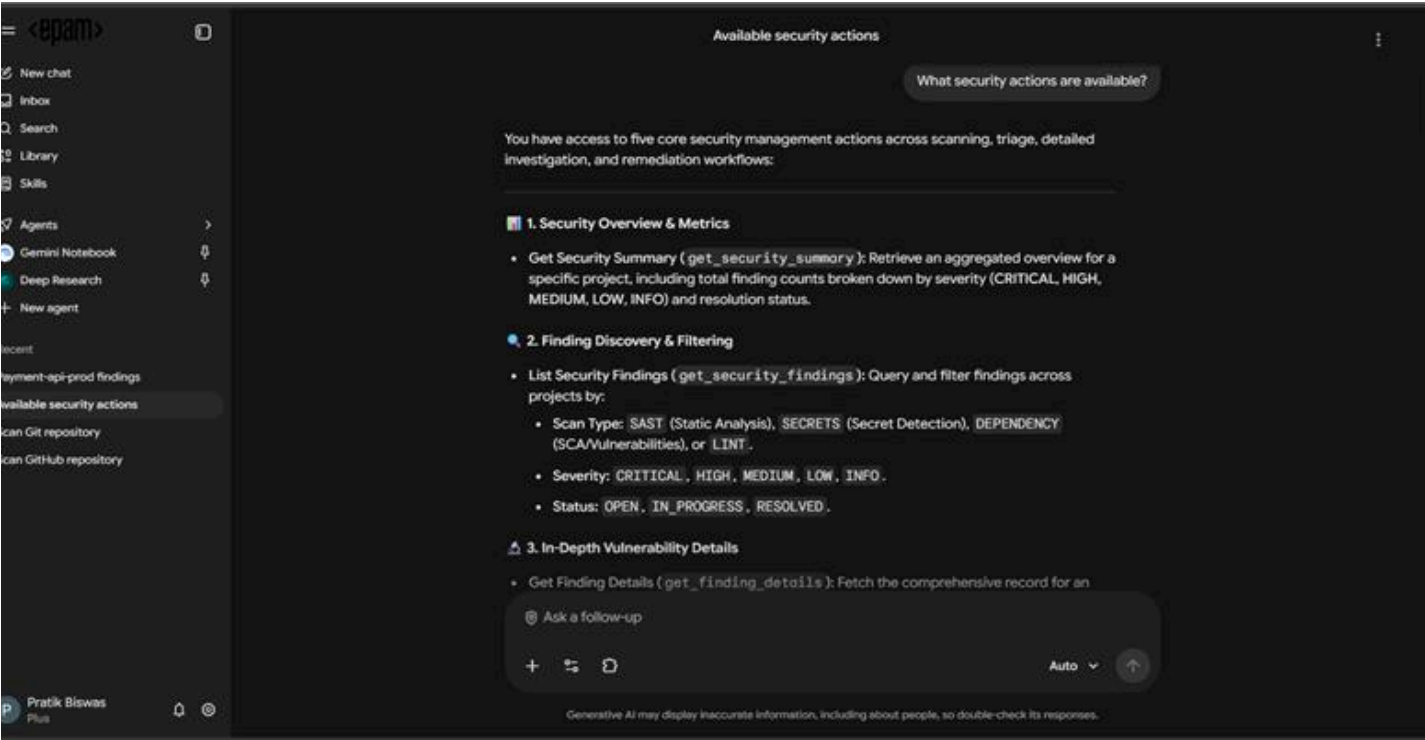
Lessons Learned

“Governed AI access is a security boundary, not just an API wrapper.”

- 01 Authentication is not optional** — Every request needs JWT validation — there’s no anonymous AI access, even for reads.
- 02 RBAC belongs in code, not prompts** — Enforcing roles at the service layer blocks bypass regardless of entry point — MCP or REST.
- 03 Audit trails must be built in** — Writing an immutable audit record inside the same transaction makes compliance the default, not an afterthought.
- 04 Two identity paths cover more ground** — Seeded demo users plus an email-role map support fast testing today and a real enterprise SSO swap tomorrow.

Visuals

GEMINI ENTERPRISE APP



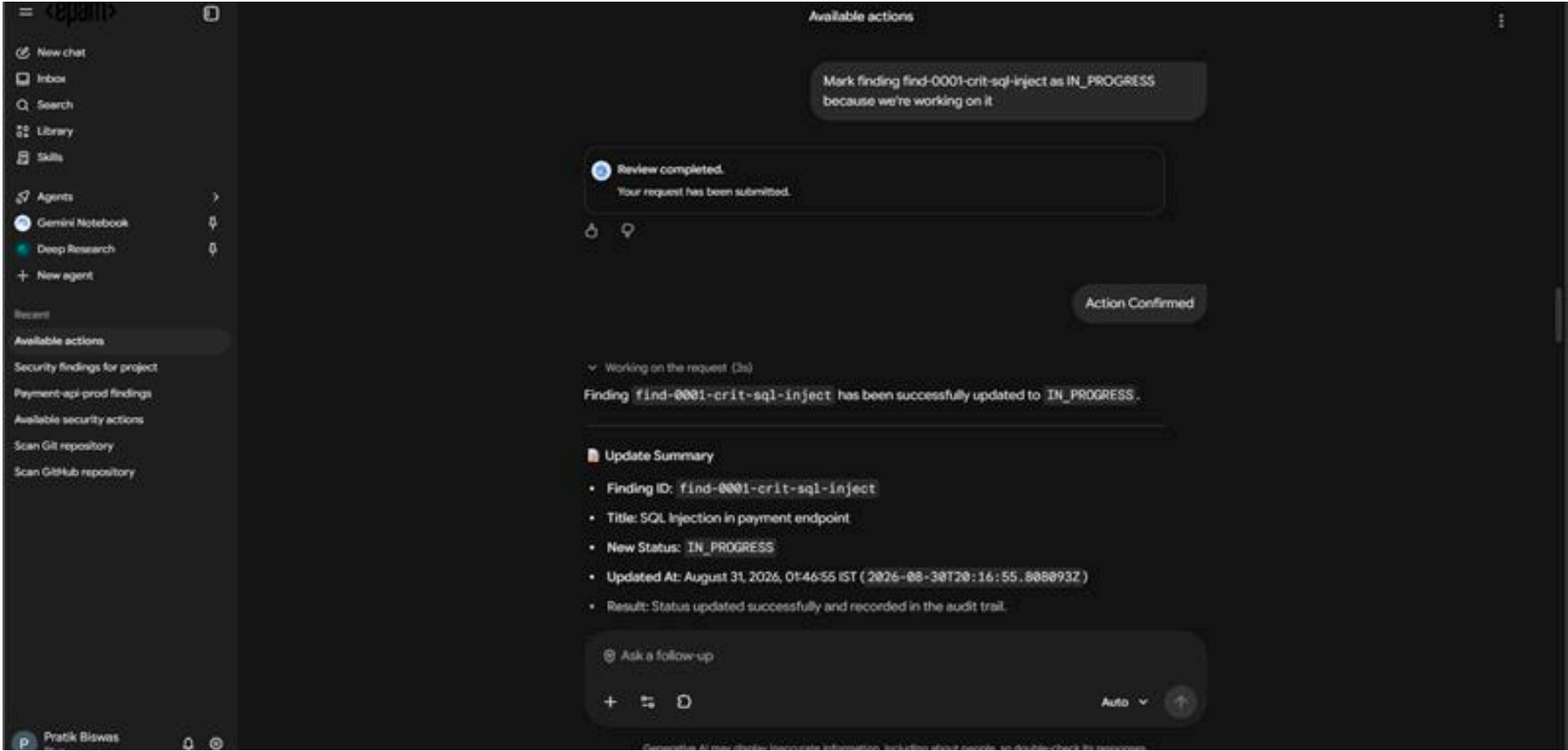
MCP CONNECTOR

⚠️ Actions for this custom MCP server are disabled. Go to the Actions tab to enable the actions you want available

SecOps Nexus MCP

Details	Entity	Alerts
Collection ID	secops-nexus-mcp_1788091964337	
Type	custom_mcp	
Region	global	
Instance URL	https://enterprise-security-connector-322397282705.us-central1.run.app/mcp	
Connected apps	gemini-enterprise-epam	
Connector state	Active	
Content policy	N/A	

DATA WRITE BACK



CUSTOM ACTIONS

Google Cloud

GCPP-CCOE-SANDBOX

Search (/) for resources, docs, products, and more

Search

Gemini Enterprise

Data

Actions New

Data stores > SecOps Nexus MCP > Actions

Actions Reload custom actions

Filter

Enter property name or value

☐	Name	Type ↓	Status
☐	Get Finding Audit History	Custom	Enabled
☐	Get Finding Details	Custom	Enabled
☐	Get Security Findings	Custom	Enabled
☐	Get Security Summary	Custom	Enabled
☐	Update Finding Status	Custom	Enabled

SECOPS NEXUS DEMO :

<https://epam.sharepoint.com/:v:/s/EPAMGeminiEnterpriseHackathon/IQD1f7UR7LErT49xk58lNWl3AeH5pmbMbLLRndPPGP2ouFco?e=dpay5p>

